

DUSTBADGE · PROBLEM STATEMENT

DustBadge problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work with sources)
0.1	2026-09-25	Amish Chadha	Initial scaffold

DustBadge problem statement

Respirable crystalline silica (RCS) causes silicosis, which is incurable, yet most exposed workers never learn their own exposure while it can still be changed. Personal sampling with a pump and filter is costly, needs a trained hygienist, and reports a single shift average days later, so a worker cannot tell during the shift that a saw, a dry sweep or a dusty haul road is putting them over the limit.

■ The problem

Silicosis killed more than 12,900 people worldwide in 2019, and the burden falls hardest on lower-income regions where protection is weakest (Chen, Liu and Xie, *BMC Pulmonary Medicine*, 2022). In the United States about 2.3 million workers are exposed, and OSHA estimated that its 2016 limit of 50 $\mu\text{g}/\text{m}^3$ (8 h average) would prevent more than 600 deaths a year once fully effective (US Department of Labor, 2016). An estimated 44 million artisanal and small-scale miners work mostly in developing economies, and a systematic review found a pooled silicosis prevalence of 23.9 % among those studied, with some disease after fewer than 6 years of mining (Howlett et al., *PLOS Global Public Health*, 2023).

Limits exist, but exposure is rarely measured where the work happens. The reference method draws air through a size-selective cyclone onto a filter, which a laboratory analyzes afterward. NIOSH has shortened the wait with its Field Analysis of Silica Tool, which gives results the same or next day using a portable infrared analyzer (NIOSH, FAST), but that still reports after the shift and needs equipment most small quarries and workshops do not have. Real-time feedback during the shift exists for US underground coal miners through the continuous personal dust monitor (US Department of Labor, 2016), a specialized instrument well beyond the budget of an informal quarry.

■ Prior work and the gap

- **Continuous personal dust monitors** (for example the Thermo PDM3700 used in US coal mines) give in-shift results for coal mine dust but are costly and not designed for silica fraction estimates in other commodities.
- **Real-time optical monitors.** NIOSH notes that factory-calibrated optical monitors can differ by up to a factor of 10 from gravimetric samples, because the signal depends on particle size, shape and refractive index; a site correction factor from a filter sample is needed (Cauda, *NIOSH Manual of Analytical Methods*, 2021).
- **Low-cost particle sensors in mining.** A calibration study found that a Plantower PMS5003 reached R^2 of 0.70 to 0.90 against a reference below 3.0 mg/m^3 of coal dust, and suggested such sensors could extend personal monitoring to all miners (Amoah et al., *Science of the Total Environment*, 2023).
- **Sensor modules with a respirable size bin.** The Sensirion SPS30 reports PM4 mass (particles up to 4 μm), close to the respirable convention, over 0 to 1,000 $\mu\text{g}/\text{m}^3$ (Sensirion SPS30 datasheet).

The gap is an open, low-cost, wearable reference design that estimates respirable dust and silica exposure through the shift, tells the worker when the shift average is heading over a limit, and keeps the worker in control of the data. No optical sensor can identify silica; DustBadge can only estimate it from respirable dust and a site-specific silica fraction, and it must say so.

■ Users and context

User	Need	Context
Quarry, pit or stone yard worker	Know during the shift when dust is high enough to act (wet the cut, move upwind, fit a respirator, stop)	Surface quarries and open pits, 8 to 12 h shifts, heat, dust, water sprays
Stone benchtop fabricator	Warning when cutting, grinding or polishing raises exposure	Small workshops, often without any exposure sampling
Construction worker	Feedback when cutting concrete, brick or tile	Building sites, short high-dust tasks
Supervisor or safety officer	Which tasks and places drive exposure, with worker consent	Site safety reviews
Occupational hygienist, researcher or worker organization	A cheap screening tool between regulatory samples; open data format	Surveys, research, community health programs

■ Constraints

- Garage-buildable prototype, about \$90 USD in parts, from off-the-shelf modules and 3D-printed parts.
- Small and light enough to wear on the chest within 30 cm of the nose and mouth for a full shift.
- Must work in heat, dust and wet spray without clogging or false alarms dominating.
- Exposure data belongs to the worker. Sharing with an employer is by consent, and the device must not become a tool for discipline.
- Research and educational prototype only. It is not certified monitoring or personal protective equipment, does not replace regulatory sampling or respirators, and is not certified for explosive atmospheres.

■ Out of scope

- Direct measurement of silica content (needs a filter and laboratory or field infrared analysis).
- Regulatory compliance sampling.
- Use in underground coal mines or any atmosphere that may contain flammable gas or combustible dust, which would require an intrinsically safe, certified design.
- Medical diagnosis or health surveillance.

■ Open questions

- First users and partner: quarry workers through a worker organization or NGO, stone fabricators, or a university occupational hygiene group. Proposed, awaiting Amish.
- Which exposure limits to use by default, given that they differ by country (for example 50 µg/m³ in the US and 100 µg/m³ as the EU binding limit). Proposed: configurable, US values as default, awaiting Amish.
- How a site silica fraction is obtained where no laboratory is near. Proposed: one filter sample per site and task analyzed by a partner laboratory, awaiting Amish.

■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate load, distance, terrain and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design